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Review

# Artificial intelligence and investment management: Structure, strategy, and governance

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## Abstract

Artificial intelligence (AI) is transforming investment management by enabling data-driven decision-making across portfolio optimization, forecasting, risk assessment, advisory services, and regulatory compliance. Despite growing interest, much of the existing research remains fragmented, focusing narrowly on specific techniques or domains. This study addresses that gap through a comprehensive and systematic literature review of 178 peer-reviewed studies indexed in Scopus, covering the structure, strategy, and governance of AI applications in investment management. Adopting established review protocols and integrating thematic and visual analysis, the review categorizes the literature into six grand themes: AI-driven portfolio management and optimization; AI in financial forecasting and market prediction; AI-powered robo-advisory and personalized financial services; AI for risk management, fraud detection, and regulatory compliance; AI and strategic investment decision-making; and enablers and constraints in AI adoption. Findings reveal that AI is not just enhancing operational efficiency but also reconfiguring strategic and institutional practices. The review identifies key research gaps related to explainability, adoption barriers, and ethical concerns, and offers an integrated framework to guide future research, policy design, and industry adoption. By synthesizing a diverse body of work, this study advances our understanding of how AI is reshaping the architecture and dynamics of modern investment management.

## Introduction

Artificial intelligence (AI) has emerged as one of the most transformative technologies of the 21st century, reshaping the contours of decision-making across industries. Defined by its ability to simulate human intelligence through machine learning, natural language processing, and advanced data analytics, AI enables systems to learn from data, identify patterns, and make autonomous decisions (International Business Machines Corporation (IBM), 2025). In the context of finance, AI is increasingly central to the evolution of investment management, where its applications include portfolio optimization, risk assessment, predictive modeling, robo-advisory services, and regulatory compliance (Finlay, 2023; Kolanovic et al., 2020). The convergence of big data availability, computational power, and algorithmic innovation has accelerated the deployment of AI tools in ass*et al*location, security selection, forecasting, and client servicing (CFA Institute Research Foundation, 2020; Alkhatib et al., 2024).

The integration of AI into investment management carries profound implications for investors, asset managers, and regulators alike. Investors gain access to more sophisticated and personalized strategies, often at reduced costs, while asset managers can enhance efficiency, scale operations, and generate more nuanced insights from vast and unstructured data (CFA Institute, 2024). At the same time, regulators must contend with the challenges posed by algorithmic opacity, systemic risk, and data ethics (Sivarajah et al., 2016; Svetlova, 2022). Despite the growing body of research on specific AI tools and their technical implementation, there remains a critical gap in synthesizing how these innovations holistically reshape investment decision-making frameworks, institutional practices, and market dynamics. This study complements existing reviews, which focus more on specific sub-domains (Aldridge, 2023; Alshater et al., 2022; Bahoo et al., 2024; Finlay, 2023; Kong et al., 2024; Lee et al., 2023), offering limited insight into the broader strategic and organizational transformations underway and associated research challenges. Unlike bibliometric and specific topic reviews, this study offers a comprehensive, concept-driven synthesis across major themes. It goes beyond tracking trends to explore how AI is transforming the structure, strategy, and governance of investment management.

This study conducts a comprehensive and systematic literature review (SLR) of peer-reviewed research articles indexed in the Scopus database that explore the intersection of AI and investment management. The review follows established SLR protocols, combining quantitative and thematic analysis to categorize 178 relevant studies published over the past decade. These studies were mapped into six grand themes based on their focus areas and conceptual alignment: AI-Driven Portfolio Management and Optimization; AI in Financial Forecasting and Market Prediction; AI-Powered Robo-Advisory and Personalized Financial Services; AI for Risk Management, Fraud Detection, and Regulatory Compliance; AI and Strategic Investment Decision-Making; and Enablers and Constraints in AI Adoption for Investment Management.

The broad findings of this review reveal that AI is not merely an operational enhancement but a strategic catalyst for transformation across all levels of the investment management ecosystem.

While some studies emphasize performance and technical sophistication, others highlight the organizational, ethical, and regulatory dimensions that determine the success or failure of AI integration. For instance, although machine learning models improve forecasting and risk analytics, their adoption is uneven across institutions due to cultural resistance, infrastructure limitations, and policy ambiguity. Similarly, while robo-advisors offer greater access and personalization, their widespread use raises questions about fiduciary responsibility, transparency, and investor behavior.

By systematically organizing and synthesizing the existing body of literature, this review delivers several contributions to the field of AI in investment management. First, it constructs a coherent thematic map that captures the wide-ranging scope and methodological heterogeneity present in current research. This includes studies spanning technical algorithm design, behavioral finance integration, institutional adoption dynamics, and regulatory implications. Second, it identifies critical gaps and forward-looking priorities—most notably in areas such as explainable AI, ethical governance frameworks, and the alignment of AI systems with the needs and expectations of diverse stakeholders including investors, asset managers, and regulators. Third, the review proposes an integrated conceptual framework that bridges academic inquiry and real-world application. This framework offers a foundation for future empirical research, supports evidence-based policy formulation, and guides strategic planning within financial institutions seeking to responsibly adopt and scale AI technologies.

Ultimately, this review advances a deeper understanding of how AI is reshaping the structural and operational foundations of modern investment management. It highlights not only the transformative potential of AI but also the critical safeguards and design principles necessary to ensure that its implementation is trustworthy, transparent, and aligned with long-term financial and societal goals. These insights are intended to inform ongoing theoretical development, practitioner strategies, and evolving regulatory discourse in the years to come.

In what follows, Section 2 outlines the systematic literature review methodology, including data sources, selection criteria, and analytical tools. Section 3 explores AI applications in portfolio optimization, focusing on deep learning, reinforcement learning, and hybrid models. Section 4 reviews AI-driven forecasting methods for asset prices, sentiment analysis, and macroeconomic trends. Section 5 examines the role of AI in robo-advisory and personalized financial services, emphasizing trust, user behavior, and system design. Section 6 discusses AI tools for risk detection, fraud prevention, and regulatory compliance, highlighting ethical and explainability concerns. Section 7 analyzes how AI supports strategic investment decisions at firm and institutional levels through scenario modeling and ESG integration. Section 8 identifies enablers and barriers to AI adoption, including organizational readiness, infrastructure, and regulatory alignment. Section 9 synthesizes cross-theme insights and implications for investors, asset managers, and regulators. Section 10 concludes the paper.

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## Section snippets

### Methodological basis

The review was initiated by formulating clearly defined research questions, each mapped to one of the six thematic domains. These questions served as the foundation for delineating the study's scope and informed a systematic approach to literature selection and analysis. A comprehensive search strategy was employed, utilizing the electronic database Scopus to capture both academic and applied contributions. Boolean logic and thematic keyword combinations were used to construct targeted queries, …

### AI-driven portfolio management optimization

This theme encompasses studies applying AI to ass*et al*location, portfolio construction, and risk-return optimization. Key topics include deep learning models such as LSTM, CNN, and DRL; reinforcement learning; portfolio optimization techniques like the Black-Litterman model and evolutionary algorithms; risk modeling; explainable AI; robo-advisors; and ESG-oriented investment strategies. …

### AI in financial forecasting and market prediction

This theme covers research leveraging AI and machine learning to forecast stock prices, mutual fund performance, market volatility, and asset returns. Key topics include hybrid deep learning models, sentiment analysis, large language models (e.g., GPT), fundamental and technical indicators, and macroeconomic forecasting. …

### AI-powered robo-advisory and personalized financial services

This theme explores AI applications in automated financial advisory systems and personalized investor services using machine learning and user profiling. Key topics include robo-advisors, investor profiling, behavioral biases, adoption models such as TAM and UTAUT, personalization, trust, and transparency. …

### AI for risk management, fraud detection, and regulatory compliance

This theme highlights the use of AI in identifying financial risks, detecting fraud, and supporting regulatory compliance. Key topics include fraud detection (e.g., credit card fraud, anti-money laundering), risk spillovers, tail risk analysis, compliance systems, and the legal and ethical use of AI. …

## AI and strategic investment decision-making

This theme focuses on the role of AI in guiding strategic investment decisions at the enterprise and policy level. Key topics include scenario analysis, macroeconomic modeling, ESG integration, resource allocation, decision-support systems, and strategic planning frameworks. …

## Enablers and constraints in AI adoption for investment management

This theme examines the factors that facilitate or hinder the adoption of AI in investment management, including technological, organizational, regulatory, and behavioral dimensions. Key topics include barriers to adoption, AI maturity, ethics, transparency, interpretability, investor attitudes, and infrastructure gaps. …

## Discussion

This systematic review reveals that artificial intelligence (AI) is reshaping the investment landscape with significant implications for investors, asset managers, and regulators. Across six thematic areas, AI is evolving from a performance-enhancing tool to a transformative force in institutional strategy, decision-making processes, and policy oversight.

Across the literature, AI optimization methods matter. While not explicitly discussed in this analysis, studies using systematic HPO, …

## Conclusions

This comprehensive review of 178 studies on the integration of artificial intelligence in investment management reveals a rapidly evolving landscape defined by six interrelated thematic domains: portfolio optimization, market forecasting, robo-advisory, risk management, strategic investment, and AI adoption enablers. These themes demonstrate that AI is no longer a peripheral innovation but a foundational force reshaping the design, execution, and oversight of financial decision-making across …

Recommended articles

## References (176)

S. Ahmed *et al.*
Artificial intelligence and machine learning in finance: A bibliometric review
Research in International Business and Finance (2022)

R.A. Al Suwaidi *et al.*
Financial literacy and FinTech market growth around the world
International Review of Financial Analysis (2024)

A. Bitetto *et al.*
Measuring financial soundness around the world: A machine learning approach
International Review of Financial Analysis (2023)

A. Bitetto *et al.*
On the efficient synthesis of short financial time series: A dynamic factor model approach
Finance Research Letters (2023)

K. Du *et al.*
Natural language processing in finance: A survey
Information Fusion (2025)

M. García-Galicia *et al.*
Continuous-time reinforcement learning approach for portfolio management with time penalization
Expert Systems with Applications (2019)

H. Hao *et al.*
Empowering scenario planning with artificial intelligence: A perspective on building smart and resilient cities
Engineering (2024)

A. Abadi *et al.*
Starlit: Privacy-preserving federated learning to enhance financial fraud detection
arXiv preprint arXiv:2401.10765 (2024)

J. Adebayo *et al.*
Sanity checks for saliency maps
Advances in Neural Information Processing Systems (2018)

M.A.M. Al Janabi

Multivariate portfolio optimization under illiquid market prospects: A review of theoretical algorithms and practical techniques for liquidity risk management
Journal of Modelling in Management (2021)

View more references

## Cited by (2)

Intelligent governance? Evidence from the Adoption of AI in Chinese a-share firms
2025, Finance Research Letters

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Organizational Dominant Logic and Strategic Diversification: The Mediating Role of BI&A Competencies
2025, Sustainability Switzerland

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